The quadratic formula

1 Considering the quadratic formula, find the values of a, b and c in the following quadratic equations:

a $x^2+7x+10=0$

b $x^2-3x-4=0$

c $2x^2+9x=0$

d $4x^2+3x=5$

e $8x^2-3x=0$

f $-8x^2+3x=0$

g $-2x^2+9x+5=0$

h $3x^2-8x+2=9x-7$

2 Solve the following equations using the quadratic formula:

a $x^2+5x+6=0$

b $x^2-5x+6=0$

c $x^2-5x+\frac{9}{4}=0$

d $x^2-8x+\frac{55}{4}=0$

e $2x^2+6x-8=0$

f $4x^2-10x+4=0$

g $4x^2+7x+3=0$

h $4x^2-17x-15=0$

i $-6+7x+5x^2=0$

j $-6-13x+5x^2=0$

k $-20-11x+3x^2=0$

l $-20+21x+5x^2=0$

3 Solve the following equations, leaving your answer in surd form:

a $x^2-7x+9=0$

b $x^2-5x-2=0$

c $-2x^2-15x-4=0$

d $3x^2+9x-4=0$

e $5x^2-15x+2=0$

f $-5x^2-15x+3=0$

g $5x=(x-5)(3x+3)$

h $\frac{3x+1}{3x-1}-\frac{3x-1}{3x+1}=5$

4 Consider the equation $2x^2=14$.

a Solve for x using the quadratic equation. Give your answer in surd form.

b Give x as decimal number correct to three decimal places.

5 Solve the following equations, rounding your answers to three decimal places:

a $x^2+3x-6=0$

b $x^2+7x-3=0$

c $4x^2+7x+2=0$

d $5x^2+9x+2=0$

e $2x(x-4)=3x+1$

f $3x(x+4)=-3x+4$

g $7.1x^2+5.3x-1.5=0$

h $1.8x^2+5.2x-2.3=0$

6 Solve the following expression for m:

$$10 - 6m + 2m^2 = m^2 + 8m + 9$$

7 For each of the given solutions to a quadratic equation:

i Find the values of a, b and c.

ii Write down the quadratic equation that has these solutions.

a $x = \dfrac{-5 \pm \sqrt{5^2 - 4 \times (-7) \times 10}}{2 \times (-7)}$

b $x = \dfrac{-22 \pm \sqrt{484 - 108}}{18}$

Applications

8 An object is launched from a height of 80 ft with an initial velocity of 107 ft/s.

After x seconds, its height, h, is given by:

$$h = -16x^2 + 107x + 80$$

Find the number of seconds, x, after which the object is 30 ft above the ground. Round your answer to one decimal place.

9 An object is launched from a height of 8 m with an initial velocity of $k\ \text{ms}^{-1}$.

After t seconds, its height, $h(t)$, is given by:

$$h(t) = -2t^2 + kt + 8$$

The object was in flight for 4 seconds before reaching the ground. What was its initial velocity k ?